

Project Cycle 3 Summary

Two-Proportion Inference & Behavioral Risk Association



Group 22



Research Question

Question 8

Does a student's emotional state (sadness/hopelessness) clearly reflect in their everyday behavior?

Group(A) : Carrot eating&No soda drinking

Group(B) : Cigarette use&Sports participation



Method

Two-Proportion Z-Test
(Independent Samples)



Conclusion

 Sadness significantly lowers the proportion of maintaining a healthy diet

 Even among sports team members, sadness significantly increases smoking rates

 Poor mental well-being creates a "dual-burden"—losing good habits while increasing risk-taking



Key Results

Group(A): Dietary Habits

Difference : -1.15%

Z-statistic : -2.21

P-value : 0.0271

95% CI : [-2.15%, -0.16%]

→ **Negative Association**

Group(B): Risk Behaviors

Difference : +2.82%

Z-statistic : 5.45

P-value: < 0.001

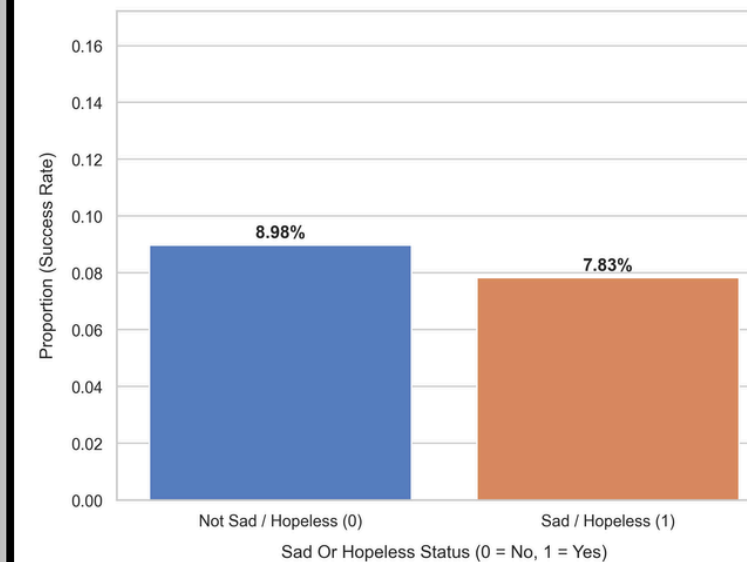
95% CI : [1.75%, 3.89%]

→ **Positive Association**

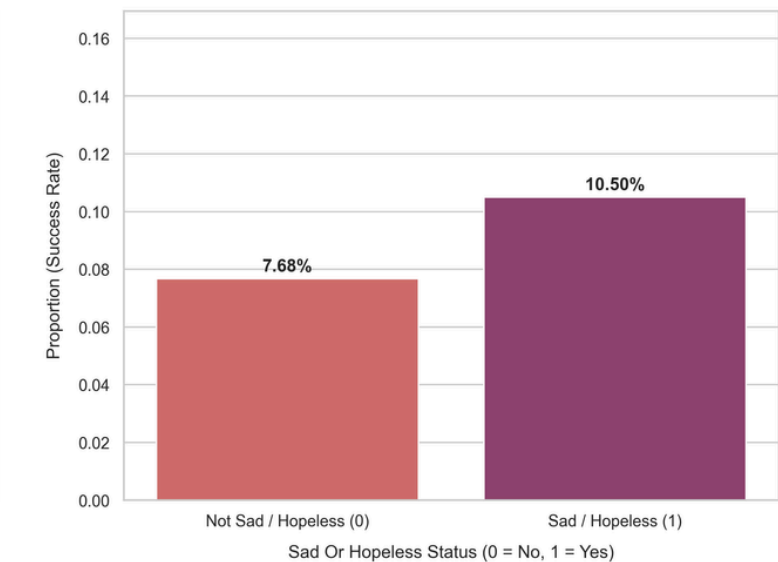


Data Visualization

Proportion of Group A (Carrot Eating & No Soda Drinking) by Mental Well-being Status



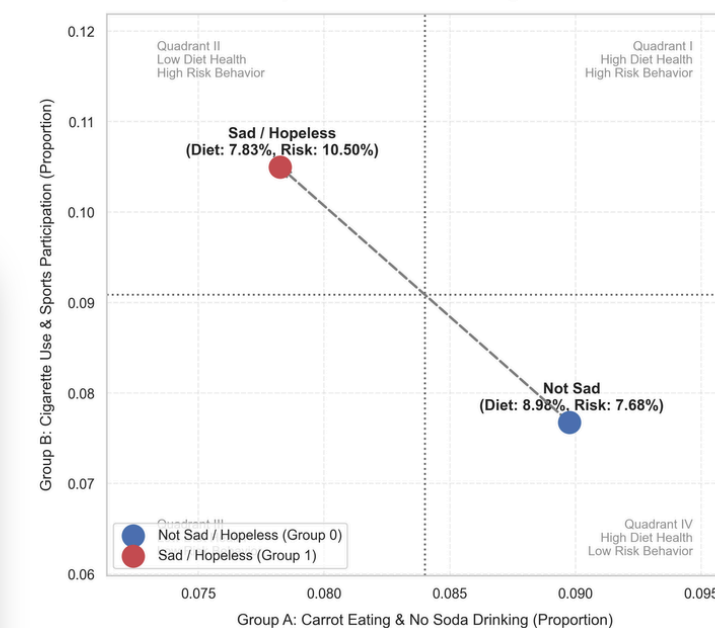
Proportion of Group B (Cigarette Use & Sports Participation) by Mental Well-being Status



 Significant Decline in Healthy Diet

 Significant Increase in Risk Behaviors

Behavioral Four-Quadrant Matrix Plot by Student Mental Well-being Status



Q2: High-Risk Zone
Low Diet, High Risk

Q4: Ideal Health Zone
High Diet, Low Risk

The dashed line proves that mental health deterioration triggers a systemic lifestyle shift from the best-case (Q4) to the worst-case scenario (Q2)